

5(a)	(energy stored =) area under line or $\frac{1}{2} QV$ $= \frac{1}{2} \times 8.0 \times 1.2 \times 10^{-4}$	C1
	$= 4.8 \times 10^{-4} \text{ J}$	A1
5(b)(i)	$(\tau =) RC$	C1
	$(\tau =) 220 \times 10^3 \times (1.2 \times 10^{-4}/8.0) = 3.3 \text{ s}$	A1
5(b)(ii)	$E \propto V^2$	C1
	(so time to) $V_o/3$ $V = V_o e^{-t/RC}$	C1
	$\frac{V_o}{3} = V_o e^{-t/3.3}$ $\frac{1}{3} = e^{-t/3.3}$	C1
	$t = 3.6 \text{ s}$	A1
5(c)	(total) capacitance is doubled	M1
	time constant is doubled	A1